# RCS02 | Instructor Guide

**CTS Cohort 6 | CTS Day 10 | 2026-07-21 | 10:00 AM-3:00 PM | 6 students**

## Residential Framing Basics: Measure, Layout, Build

**Core rule:** A professional does not guess. A professional checks, protects, and verifies.

**Big idea:** Accurate framing begins before the saw starts: read the plan, measure twice, mark clearly, and verify the cut list.

## Architecture of the day

| Time | Instructor move |
|---|---|
| 10:00-10:15 | Arrival, PPE, retrieval warm-up |
| 10:15-10:30 | Morning quiz |
| 10:30-10:50 | Quiz review and measurement misconception check |
| 10:50-11:20 | Tape-measure and fraction model |
| 11:20-12:00 | Wall-member map and window-framing demonstration |
| 12:00-12:30 | Cut-list guided practice and lab release |
| 12:30-1:00 | Lunch and controlled reset |
| 1:00-2:25 | Individual capstone framing lab |
| 2:25-2:45 | Dimensions, diagonal QC, evidence, cleanup |
| 2:45-3:00 | Exit ticket and Day 3 preview |

## Learning targets

- Read a tape measure to 1/16 inch and add common fractions.
- Identify plates, studs, king studs, jack studs, header, sill, and cripples.
- Lay out a 4-foot by 3-foot training wall with a framed window opening.
- Complete an instructor-approved cut list and assemble a square wall frame.

## Materials and setup

- Six approved 4-foot by 3-foot capstone wall drawings and lumber kits
- Tape measures, pencils, squares, clamps, and member labels
- Instructor-controlled cutting tools with guards and task dust controls
- Approved fasteners and drivers
- Eye/hearing protection and station-control signage
- Evidence camera or approved device

## Instructor talking points

- Connect RCS01 systems thinking to structure: every later system depends on an accurate frame.
- Model tape-measure reading, fraction addition, a clear carpenter's mark, and a two-person verification callout.
- Teach the load-path idea at training level: roof/floor loads move through framing to the foundation.
- Explain the window-opening members and why dimensions must follow the approved training drawing.
- Demonstrate inspection of lumber, tool guards, work-zone control, clamping, and safe body position.

## Whiteboard plan

1. Core rule
2. 1/16-inch ladder
3. Wall member sketch
4. Measure -> mark -> verify -> cut
5. QC: dimensions, square, fasteners, labels

## Lab facilitation

Each student begins an individual 4-foot by 3-foot wall capstone with a framed window opening. Students inventory stock, complete the approved cut list, mark and verify all members, make supervised cuts only after clearance, dry-fit, fasten, and complete a square/dimension inspection.

### Lab sequence

6. PPE and station inspection
7. Read training drawing
8. Create cut list
9. Mark and partner-verify
10. Instructor cut authorization
11. Supervised cutting
12. Dry layout and member labels
13. Fasten frame
14. Measure diagonals and correct
15. Photograph, document, reset

### Safety boundaries

- Eye and hearing protection in active cutting zones
- Instructor authorization before each cutting operation
- Guards stay installed and functional

- Clamp or support stock
- Wall stays flat until instructor authorizes movement
- Stop work for conflicting measurements or damaged material

### Evidence targets

- Completed measurement sheet
- Approved cut list
- Labeled dry layout photo
- Final wall dimensions and diagonal check
- Instructor tool-control observation

## Quiz answer key

**1. What is the smallest graduation students must read today?** 1/16 inch.

**2. What member runs horizontally at the bottom of a framed wall?** Bottom plate.

**3. What stud directly supports a header?** Jack or trimmer stud.

**4. What stud runs full height beside the jack stud?** King stud.

**5. Why measure diagonals?** To check whether the frame is square.

**6. What must happen before any cut?** The mark and cut list are verified and the instructor authorizes the cut.

**7. What does on-center describe?** Spacing measured from the center of one framing member to the center of the next.

**8. Name one reason lumber is inspected before use.** To find damage, severe warp, splits, embedded metal, or another unsafe defect.

**9. What is the day's core professional habit?** Check, protect, and verify rather than guess.

**10. When may a student bypass a tool guard?** Never.

## Exit-ticket look-fors

- One measurement skill I can now demonstrate
- One wall member and its job
- One verification I completed
- One step I need to practice

## Google Classroom post

RCS Day 2 - Residential Framing Basics: Measure, Layout, Build

Today we focused on accurate framing begins before the saw starts: read the plan, measure twice, mark clearly, and verify the cut list. Students completed guided instruction, hands-on capstone

work, verification, evidence capture, and cleanup. Submit your worksheet, lab evidence, and exit ticket before leaving.

## Source and safety references

- **Canonical RCS skeleton:** /CANONICAL/00_INBOX/Downloads/rcs_curriculum_plan.md
- **Cohort 6 RCS01 source:** /CANONICAL/02_PROJECTS/CTS_Cohort_06/03_DAILY/D09_RCS01_2026-07-20/Source/lesson.md
- **OSHA PPE:** https://www.osha.gov/personal-protective-equipment/construction
- **OSHA eye and face protection:** https://www.osha.gov/laws-regs/regulations/standardnumber/1926/1926.102
- **OSHA circuit lockout/tagging:** https://www.osha.gov/laws-regs/regulations/standardnumber/1926/1926.417
- **Oregon electrical codes:** https://www.oregon.gov/bcd/codes-stand/pages/electrical.aspx
- **Oregon permits:** https://www.oregon.gov/bcd/lbdd/pages/oregon-permits.aspx

# RCS02 | Morning Quiz - Answer Key

**CTS Cohort 6 | CTS Day 10 | 2026-07-21 | 10:00 AM-3:00 PM | 6 students**

## Residential Framing Basics: Measure, Layout, Build

**1. What is the smallest graduation students must read today?**

1/16 inch.

**2. What member runs horizontally at the bottom of a framed wall?**

Bottom plate.

**3. What stud directly supports a header?**

Jack or trimmer stud.

**4. What stud runs full height beside the jack stud?**

King stud.

**5. Why measure diagonals?**

To check whether the frame is square.

**6. What must happen before any cut?**

The mark and cut list are verified and the instructor authorizes the cut.

**7. What does on-center describe?**

Spacing measured from the center of one framing member to the center of the next.

**8. Name one reason lumber is inspected before use.**

To find damage, severe warp, splits, embedded metal, or another unsafe defect.

**9. What is the day's core professional habit?**

Check, protect, and verify rather than guess.

**10. When may a student bypass a tool guard?**

Never.

## Scoring guidance

One point per item. Accept technically equivalent wording. Use errors to reteach before lab release. Safety-boundary errors must be corrected explicitly before the student begins the related task.

# RCS02 | Student Packet

**CTS Cohort 6 | CTS Day 10 | 2026-07-21 | 10:00 AM-3:00 PM | 6 students**

## Residential Framing Basics: Measure, Layout, Build

**Core rule:** A professional does not guess. A professional checks, protects, and verifies.

**Big idea:** Accurate framing begins before the saw starts: read the plan, measure twice, mark clearly, and verify the cut list.

## Learning targets

- Read a tape measure to 1/16 inch and add common fractions.
- Identify plates, studs, king studs, jack studs, header, sill, and cripples.
- Lay out a 4-foot by 3-foot training wall with a framed window opening.
- Complete an instructor-approved cut list and assemble a square wall frame.

## Today's run

10:00-10:15 | Arrival, PPE, retrieval warm-up

10:15-10:30 | Morning quiz

10:30-10:50 | Quiz review and measurement misconception check

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12:00-12:30 | Cut-list guided practice and lab release

12:30-1:00 | Lunch and controlled reset

1:00-2:25 | Individual capstone framing lab

2:25-2:45 | Dimensions, diagonal QC, evidence, cleanup

2:45-3:00 | Exit ticket and Day 3 preview

## Key words

top plate | bottom plate | stud | king stud | jack stud | header | sill | cripple | on-center | crown | square

## Guided worksheet

1. Convert: 1/2 = __/16; 3/4 = __/16; 7/8 = __/16.

_____

_____

2. Add: 14 3/8 + 9 7/16. Show the denominator step.

_____

_____

3. Label the wall members on the instructor-provided drawing.

_____

_____

4. Write the cut-list quantity, length, and purpose for each member.

_____

_____

5. Record two diagonal measurements and explain what they show about square.

_____

_____

## Afternoon lab mission

Each student begins an individual 4-foot by 3-foot wall capstone with a framed window opening. Students inventory stock, complete the approved cut list, mark and verify all members, make supervised cuts only after clearance, dry-fit, fasten, and complete a square/dimension inspection.

### Safety boundary

- Eye and hearing protection in active cutting zones
- Instructor authorization before each cutting operation
- Guards stay installed and functional
- Clamp or support stock
- Wall stays flat until instructor authorizes movement
- Stop work for conflicting measurements or damaged material

### Build / test sequence

[ ] PPE and station inspection

[ ] Read training drawing

[ ] Create cut list

[ ] Mark and partner-verify

[ ] Instructor cut authorization

[ ] Supervised cutting

[ ] Dry layout and member labels

[ ] Fasten frame

[ ] Measure diagonals and correct

[ ] Photograph, document, reset

## Evidence record

[ ] Completed measurement sheet: _____

[ ] Approved cut list: _____

[ ] Labeled dry layout photo: _____

[ ] Final wall dimensions and diagonal check: _____

[ ] Instructor tool-control observation: _____

## Reflection and exit ticket

One measurement skill I can now demonstrate

_____

_____

One wall member and its job

_____

_____

One verification I completed

_____

_____

One step I need to practice

_____

_____

## Instructor verification

Status: [ ] Approved [ ] Correct and recheck [ ] Needs follow-up

Instructor initials: _____ Date/time: _____

# RCS02 | Morning Quiz

**CTS Cohort 6 | CTS Day 10 | 2026-07-21 | 10:00 AM-3:00 PM | 6 students**

## Residential Framing Basics: Measure, Layout, Build

Name: _____ Date: _____

Answer from your own understanding. This is a teaching check, not a trap.

**1. What is the smallest graduation students must read today?**

_____

_____

**2. What member runs horizontally at the bottom of a framed wall?**

_____

_____

**3. What stud directly supports a header?**

_____

_____

**4. What stud runs full height beside the jack stud?**

_____

_____

**5. Why measure diagonals?**

_____

_____

**6. What must happen before any cut?**

_____

_____

**7. What does on-center describe?**

_____

_____

**8. Name one reason lumber is inspected before use.**

_____

_____

**9. What is the day's core professional habit?**

_____

_____

**10. When may a student bypass a tool guard?**

_____

_____